F16AG421-VS
VARIABLE SPEED DRIVE

11,000V
RD WH BU
FCF 100A 100A 100A
QAC 400A
BCT1 50 / 1A 2.5VA 5P20
BCT2 50 / 1A 2.5VA 5P20
BCT3 50 / 1A 2.5VA 5P20
BCN 50 / 1A 1.5VA 10P10
BAD
PFV1 RD RD RD
QCE 400A
EARTH BAR
X4-1 X4-2 X4-3 X4-4 X4-5 X4-6 X4-7 X4-8 X4-11 X4-12 X4-13
2000-EL003-F16-10016 SHEET 14

L1 L2 L3
LINE SIDE INCOMING SUPPLY SHEET 6

SYSTEM INTERCONNECTION DIAGRAM 2000-EL005-B02-10001

OUTGOING MOTOR SUPPLY SHEET 6
T1 T2 T3
CABLE SCREEN EARTHED
M 3~
F16-AG-421
ROUGHER FLOTATION CELL 1 AGITATOR
3,300V
500 kW

F16-AG-421
MOTOR HEATER TERMINAL BOX
F16-HT-541
X2 54 53
F16HT541-L F16HT541-N
0.3kW
ANTI-CONDENSATION HEATER

MOTOR SPACE HEATER CIRCUIT SHEET 6

X81 (PART OF)
RTD TEMPERATURE MONITOR RELAY
SHEET 6

F16-AG-421
MOTOR RTD TERMINAL BOX

PT100 RTD INPUT CH1	+ 13 / COM 14 / - 15	F16AG421A-RD / F16AG421A-WH / F16AG421A-BK	MOTOR WINDING RTD (RD PHASE) (TD1)
PT100 RTD INPUT CH2	+ 16 / COM 17 / - 18	F16AG421B-RD / F16AG421B-WH / F16AG421B-BK	MOTOR WINDING RTD (WH PHASE) (TD3)
PT100 RTD INPUT CH3	+ 19 / COM 20 / - 21	F16AG421C-RD / F16AG421C-WH / F16AG421C-BK	MOTOR WINDING RTD (BU PHASE) (TD5)
PT100 RTD INPUT CH4	+ 22 / COM 23 / - 24	F16AG421D-RD / F16AG421D-WH / F16AG421D-BK	MOTOR BEARING RTD (DRIVE END) (BD10)
PT100 RTD INPUT CH5	+ 25 / COM 26 / - 27	F16AG421E-RD / F16AG421E-WH / F16AG421E-BK	MOTOR BEARING RTD (NON DRIVE END) (BD20)
PT100 RTD INPUT CH6	+ 28 / COM 29 / - 30		
PT100 RTD INPUT CH7	+ 31 / COM 32 / - 33		
PT100 RTD INPUT CH8	+ 34 / COM 35 / - 36		

RTD

F16-SW-011 11kV FEEDER START/STOP CONTROL 2000-EL003-F16-10016 SHEET 7 3A & 3B
X3 3 1139 X2 67
X3 17 1383 X2 68 K41 K41
MV CONTACTOR ENABLE CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER VSD HEALTHY 2000-EL003-F16-10016 SHEET 10 4A
X2 4 1179 X2 63 K41 K41 K41
X2 25 1199 X2 64
MV INPUT ENABLE CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER VSD TRIP 2000-EL003-F16-10016 SHEET 10 3A
X2 3 1219 X2 65
X2 24 1239 X2 66 K41 K41 K41
MV INPUT TRIP CIRCUIT SHEET 6

F16-SW-011 11kV FEEDER ARC FLASH TRIP 2000-EL003-F16-10016 SHEET 8 5A & 5D
X2 56 1259 X2 70
X2 44 1279 X2 69
A30 (PART OF) X2 15 TRIP X2 16
ARC FLASH DETECTION TRIP SIGNAL SHEET 6

F16-DB-101
2000-F16-TLD-EL-10054
RD WH BU BK
R5 W5 B5 25A
EARTH BAR
415VAC
F16DB101-R5 F16DB101-W5 F16DB101-B5 F16DB101-BK5
X4 1 2 3 N PE1
VSD AUXILIARY POWER SUPPLY SHEET 6

F16-DB-201
2000-F16-TLD-EL-10057
RD BK
R1 10A
EARTH BAR
240VAC
F16DB201-R1 F16DB201-BK1
X4 5 6 PE2
VSD UPS POWER SUPPLY SHEET 6

TO FLOTATION 11kV SWITCHROOM
F16-COM-001 COMMUNICATION RACK
MODBUS ETHERNET SWITCH
F16-ETH-001
2000-F16-TER-IC-10002
CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6 — CAT6
X47 ETHERNET ADAPTOR RJ45
A2 (PART OF) DI-2B 9
X2 13 14
24VDC CONTROL SHEET 6

SAFE TORQUE OFF CIRCUIT SHEET 6
X86 1 2 3 4
F16AG421VS-1409 F16AG421VS-1419 F16AG421VS-1429 F16AG421VS-1439

F16AG421-LCS
LOCAL CONTROL STATION
X1 1 2 3 4 LOS
X1 5 6 E START
F16AG421VS-1459 F16AG421VS-1469

NOTES

1. WIRE NUMBER FERRULES ON TERMINAL STRIP SHALL BE PREFIXED WITH EQUIPMENT NUMBER e.g. F16AG421-1200, UNLESS OTHERWISE NOTED.
2. EACH WIRE SHALL BE IDENTIFIED AT BOTH ENDS WITH FERRULES.

LEGEND

- 11kV SWITCHBOARD TERMINAL
- VSD TERMINAL
- FIELD TERMINAL
- DCS OR I/O TERMINAL
- MARSHALLING TERMINAL
- DEVICE TERMINAL

PANEL A02

2000-F16-SCM-EL-10103_Rev1

F16-SW-011 F16-AG-421 rougher flot. cell 1 agitator VSD feeder - schematic diagram

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\ELVF16 - ROUGHER & SCAVENGER FLOTATION\2000-F16-SCM-EL-10103.DWG 09SEP25 6:38AM HOTSEAT04